

Case Report

Efficacy of coconut coir composite filters for oil and grease removal: A qualitative ATR-FTIR pilot study

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ABSTRACT

Wastewater from restaurants and workshops containing high concentrations of oil and grease (O&G) poses a significant environmental challenge, necessitating effective and accessible treatment methods. This qualitative, proof-of-concept pilot study investigated the efficacy of low-cost, locally sourced filter materials—coconut coir, recycled beach sand, activated charcoal, and inactivated charcoal—in various combinations for removing O&G. Filtration prototypes were developed and tested on real wastewater samples. Attenuated Total Reflectance-Fourier Transform Infrared (ATR-FTIR) spectroscopy was employed to characterise the chemical changes pre- and post-filtration. Results indicated that a combination of coconut coir, recycled beach sand, and activated charcoal (M3) was identified as the most effective for oil removal from restaurant wastewater, evidenced by the disappearance of the ester carbonyl peak (1744 cm^{-1}) and other oil-related aliphatic C–H stretches. For grease removal from workshop wastewater, a combination of coconut coir, recycled beach sand, and inactivated charcoal (G4) demonstrated superior performance, marked by the absence of key aliphatic C–H stretches (2922 cm^{-1} , 2853 cm^{-1}) characteristic of grease. In both optimal filter combinations, an increased intensity of hydroxyl (O–H) group signals in the filtrate was observed, correlating with effective O&G removal. This research highlights the potential of these sustainable filter systems for decentralized wastewater treatment, contributing to responsible resource utilisation (SDG 12). The findings offer a practical framework for designing pollutant-specific, low-cost filtration systems for small-scale enterprises.

1. Introduction

The pervasive issue of water contamination by oil and grease (O&G) from industrial and domestic effluents represents a critical global environmental challenge, demanding urgent and sustainable remediation strategies [1]. Wastewater originating from establishments such as restaurants and automotive workshops is frequently characterized by high concentrations of O&G, which pose substantial threats to aquatic ecosystems and the operational efficiency of wastewater treatment infrastructure [2]. These pollutants are known for their persistence and tendency to cause blockages in drainage systems, generate malodorous compounds, and induce corrosion in sewer lines. Furthermore, the presence of significant O&G loads in municipal wastewater treatment plants can lead to the formation of resilient scum layers, impede filtration processes, and ultimately contaminate the final sludge, thereby complicating its safe disposal and management [3]. Consequently, the

development of effective and economically viable methods for O&G removal at or near the source is paramount to mitigate environmental pollution and safeguard public health.

Conventional approaches for O&G separation, including chemical coagulation, dissolved air flotation, and membrane filtration, often achieve high removal efficiencies but entail significant operational costs, intricate process controls, and the generation of secondary waste streams [1]. For instance, while membrane technologies can yield excellent effluent quality, they are frequently limited by severe fouling when processing high lipid concentrations [4]. Additionally, while biological treatments utilising specific bacterial consortia offer an eco-friendly mechanism for degrading hydrocarbon pollutants [5], they often require stringent environmental controls and extended retention times, which can be impractical for decentralized, high-load point sources. This has spurred considerable research interest in exploring cost-effective, eco-friendly, and readily available alternative materials

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for O&G adsorption. In this context, the utilisation of natural organic fibers and agricultural byproducts, such as coconut coir, has emerged as particularly promising. These materials offer inherent biodegradability, low cost, and widespread availability, addressing the dual challenge of pollutant removal and sustainable waste management, as the disposal of such agricultural residues can itself be problematic and costly [6]. Coconut coir, the fibrous material extracted from the husk of *Cocos nucifera* L., possesses a unique lignocellulosic composition and a highly porous structure, rendering it an effective natural adsorbent for various pollutants, including O&G [7].

While the individual adsorptive capacities of materials like coconut coir, sand, and charcoal have been documented [8], there is a compelling need to investigate the synergistic performance of composite filter systems tailored for specific O&G-laden wastewater streams, such as those from restaurants and workshops. Characterizing the changes in the chemical fingerprint of wastewater pre- and post-treatment using techniques like Attenuated Total Reflectance-Fourier Transform Infrared (ATR-FTIR) spectroscopy is crucial for understanding the underlying removal mechanisms and optimizing filter design.

This study addresses a critical gap by systematically evaluating composite filter systems made from readily available, low-cost materials—coconut coir, recycled beach sand, and both activated and inactivated charcoal. The primary novelty of this work lies not in the materials themselves, but in the direct, comparative assessment of tailored composite filters for two distinct, real-world O&G waste streams: oleochemical-rich restaurant wastewater versus mineral/synthetic-based workshop grease. This approach moves beyond generic adsorbent testing to investigate pollutant-specific efficacy, addressing a key practical gap in decentralized wastewater treatment. The primary novelty of this work is the direct evaluation of multi-component filters tailored specifically for complex, real-world effluents from restaurants (oil-focused) and workshops (grease-focused), and directly comparing activated and inactivated charcoal within these composites to determine the optimal adsorbent for different types of O&G waste streams. Through detailed ATR-FTIR analysis, this research elucidates the functional group alterations indicative of O&G removal, thereby providing insights into the adsorption mechanisms.

This research contributes to the development of practical, decentralized wastewater treatment solutions. The use of waste materials aligns with SDG 12 (Responsible Consumption and Production) by promoting a circular economy, as highlighted in recent sustainability analyses of water treatment technologies ([9]; [10]). The findings offer scalable, evidence-based solutions for small-scale enterprises and communities with limited access to advanced treatment facilities, thereby fostering environmental health and sustainability.

2. Methodology

This section details the systematic approach undertaken to evaluate the efficacy of various filter combinations for O&G removal, encompassing material preparation, prototype development, wastewater sampling, and analytical characterization. The primary filter materials investigated were coconut coir, recycled beach sand, activated coconut shell charcoal, and inactivated coconut shell charcoal. A consistent weight of 400 g was used for each filter material within the filtration columns.

Mature coconut husks were sourced locally in order to obtain the coconut coir. The coir was manually extracted from the husks using a mortar and pestle to separate it from the coarser fibers. The extracted coir was then thoroughly washed with distilled water to eliminate adhering impurities. Subsequently, the cleaned coir was oven-dried at 60–80 °C for 24 hours, following the procedure outlined by Abdul Hamid et al. [11]. The dried coir was then cut and ground to achieve a finer particle size and sieved using fine woven wire sieves (ranging from 20 µm to 1.25 mm) to ensure particle size homogeneity [12].

Beach sand was collected from a local beach. To remove impurities

and fine silt particles, the sand was washed extensively with distilled water (8–10 times) until the wash water was clear. Subsequently, the sand was oven-dried at 105 °C for 24 hours to ensure sterilization and remove all moisture, preventing any potential microbial contamination from influencing the results. This pre-treatment was crucial to ensure the suitability of the sand as a filter medium without negatively impacting the clarity of the treated water. Granular activated and inactivated charcoal derived from coconut shells were procured from a commercial supplier. These materials were used as received.

The filtration prototypes were designed as modular, multi-stage column filters to assess the performance of different material combinations (see Fig. 1). The main body of each filtration unit was constructed using white PVC pipes (3 feet length, 4 inches diameter). Five PVC pipe covers (4 inches diameter), five pairs of PVC pipe connectors (2 cm diameter), and a 50 cm length of grey PVC pipe (2 cm diameter) were used for assembling the filter columns and connections. For structural stability, a 200 cm length of grey PVC pipe (2 cm diameter) and eight L-shaped connectors (2 cm diameter) were used to build a supporting base frame. Ancillary tools included a small hand saw, white sealing tape, a hot glue gun, soldering equipment, iron wire, fine mesh tea filters, and scissors. Safety equipment, including cloth gloves and face masks, were utilized during construction, particularly during soldering to prevent injury and inhalation of fumes.

The filtration system was constructed using 4-inch white PVC pipes serving as the primary housing. To create inlet and outlet ports, a 2.5 cm diameter aperture was fabricated in the center of each pipe cover via thermal perforation. To ensure uniform hydraulic distribution, a central support conduit—referred to as the “backbone”—was fashioned from 2 cm diameter grey PVC piping cut to 25 cm lengths. Longitudinal slits were machined along these conduits to facilitate radial water flow through the filter media.

The system components were assembled by fitting 2 cm pipe connectors into the prepared cover apertures, utilising polytetrafluoroethylene (PTFE) thread seal tape to ensure watertight integrity. A fine mesh screen was positioned within the inter-stage connectors to retain filter media and prevent particle migration. The entire vertical assembly was mounted on a structural frame composed of 200 cm grey PVC piping and 90-degree elbow fittings, reinforced with galvanized wire for stability.

The prototype consisted of three vertical stages, each packed with 400 g of compacted filter media. The stratigraphic configuration comprised coconut coir in the basal stage, activated or inactivated charcoal in the intermediate stage, and recycled beach sand in the uppermost stage. Wastewater was introduced into the column via gravity flow. While hydraulic retention time and flow rate were not actively controlled in this preliminary investigation, they are noted as critical variables for future optimization. Effluent was collected in a basin positioned directly beneath the apparatus.

2.1. Wastewater sample collection

Wastewater samples were collected from the drains of three distinct restaurants and three workshops located in the Kubang Kerian area, Kelantan, to obtain representative O&G-laden effluents. Clean 500 ml glass bottles were used for sample collection. Plastic bottles were strictly avoided to prevent the potential leaching of plasticizers, which can interfere with the spectroscopic analysis of organic compounds [13]. For analyses requiring larger volumes, particularly for low O&G concentrations or specific hydrocarbon analyses, 1000 ml samples were collected as per standard guidelines. Bottle caps were equipped with inert inserts to prevent contamination from the cap material.

The filter combinations for restaurant wastewater (oil removal) were designated M1 (coconut coir only), M2 (coconut coir + recycled beach sand), M3 (coconut coir + recycled beach sand + activated charcoal), and M4 (coconut coir + recycled beach sand + inactivated charcoal). For workshop wastewater (grease removal), the combinations were G1,

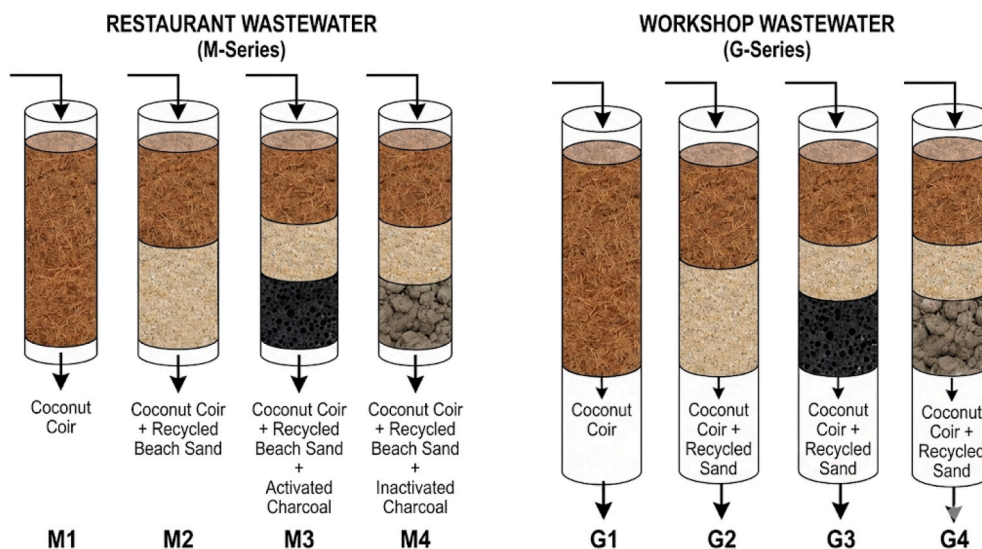


Fig. 1. Schematic diagram of the 3-stage modular vertical column.

G2, G3, and G4, following the same material sequence as illustrated below.

2.2. Functional characterization using ATR-FTIR spectroscopy

To investigate the chemical changes in the wastewater before and after filtration and to identify the functional groups involved in O&G removal, Attenuated Total Reflectance-Fourier Transform Infrared (ATR-FTIR) spectroscopy was employed. A Thermo Scientific ATR-FTIR spectrometer equipped with a Zinc Selenide (ZnSe) crystal was used. A small drop of the liquid wastewater sample (both raw and filtered) was placed directly onto the ZnSe crystal. The samples were scanned in the mid-infrared region from 4000 cm^{-1} to 600 cm^{-1} with a spectral resolution of 4 cm^{-1} and an accumulation of 32 scans per sample. Each sample was analyzed in triplicate to ensure the reproducibility of the spectral data. The resulting spectra provided information on the presence, absence, or changes in intensity of characteristic absorption bands corresponding to various functional groups in the O&G components.

2.3. Data analysis

The triplicate spectra for each sample were baseline-corrected and averaged to produce a representative spectrum for analysis. The investigation focused on the qualitative changes in the presence, absence, and relative intensity of characteristic functional group peaks to infer the removal of O&G components. Because this study was designed as a qualitative, proof-of-concept investigation aimed at mechanistic exploration via spectroscopy, quantitative statistical tests were not performed.

3. Results

This section presents the findings from the ATR-FTIR spectroscopic analysis of wastewater samples from restaurant and workshop drains, before and after filtration with different material combinations.

3.1. ATR-FTIR characterization of unfiltered wastewater

The average ATR-FTIR spectrum of unfiltered restaurant wastewater as per following Figure exhibited characteristic peaks indicative of oil components. Prominent narrow peaks at 2922 cm^{-1} and 2855 cm^{-1} were attributed to asymmetric and symmetric C-H stretching of methyl and methylene groups in aliphatic chains, respectively, consistent with

long-chain carbon structures found in lipids [14,15].

A band at 3006 cm^{-1} corresponded to = C-H (*cis*-) stretching. A strong, narrow peak at 1744 cm^{-1} was assigned to the ester carbonyl (C=O) functional group of triglycerides, validating the presence of vegetable oils and animal fats. We have updated our spectral baseline justifications to align with recent literature (e.g., removing pre-2015 citations and retaining recent functional group validations such as [16,17]) to ensure the peak assignments accurately reflect current spectroscopic consensus for wastewater profiling. Other notable absorptions included C-H bending of *cis*-disubstituted olefins at 1418 cm^{-1} , CH_3 methyl group bending/rocking at 1377 cm^{-1} [18], C-O stretching of ester groups around $1160\text{--}1097\text{ cm}^{-1}$ [19], and CH_2 rocking vibrations overlapping with out-of-plane C-H stretching of mono-substituted olefins at 722 cm^{-1} [20].

The average ATR-FTIR spectrum of unfiltered workshop wastewater as per the following figure indicate distinct features associated with grease. A broad peak around 3381 cm^{-1} indicated the presence of O-H (hydroxyl) groups [21].

Strong absorbances in the $2954\text{--}2853\text{ cm}^{-1}$ region were characteristic of asymmetric and symmetric vibrations of methyl (CH_3) and methylene (CH_2) hydrophobic groups, likely originating from components like lithium 12-hydroxystearate often found in greases [21]. Specifically, CH_3 asymmetric vibrations appeared at 2954 cm^{-1} , while CH_2 asymmetric and symmetric stretching occurred at 2922 cm^{-1} and 2853 cm^{-1} , respectively. An intense peak at 1634 cm^{-1} was assigned to the Amide I band (C=O stretching), suggesting the presence of proteinaceous or amide-containing compounds [17]. The CH_3 bending/rocking vibration was again observed at 1377 cm^{-1} . Bands around 720 cm^{-1} were attributed to = C-H bending of *cis*-disubstituted alkenes and rocking vibrations of methylene groups [22].

3.2. ATR-FTIR characterization of filtered wastewater

Post-filtration spectra in the following figure revealed significant changes in Restaurant Wastewater (Oil Removal). A broad band between 3700 and 3000 cm^{-1} , corresponding to -OH vibrations from alcohols, phenols, and carboxylic acids, showed the highest intensity for samples filtered through M3 (coconut coir + recycled beach sand + activated charcoal), while M1 (coconut coir only) exhibited the weakest intensity in this region.

A sharp band around $1635\text{--}1636\text{ cm}^{-1}$, attributed to C-H bonds of methyl and methylene groups of lipids, was most intense in the M3 filtrate and least intense in the M1 filtrate.

Critically, the characteristic ester carbonyl peak ($\text{C}=\text{O}$ stretch) at 1744 cm^{-1} , prominent in the unfiltered sample (M0), was absent in all filtered samples (M1, M2, M3, M4), as summarized in following Table. Other changes included the absence of the $\text{C}-\text{H}$ stretch at 2872 cm^{-1} and CH_3 bending at 1377 cm^{-1} post-filtration. Conversely, new broad peaks corresponding to $\text{O}-\text{H}$ groups and $\text{C}-\text{O}$ from environmental CO_2 appeared (Fig. 5).

Filtered workshop wastewater spectra in the above figure also demonstrated notable alterations. Similar to oil filtration, the broad $\text{O}-\text{H}$ band ($3700\text{--}3000\text{ cm}^{-1}$) was most intense for samples filtered with G4 (coconut coir + recycled beach sand + inactivated charcoal) and weakest for G1 (coconut coir only). The peak at 1636 cm^{-1} , typically associated with the $\text{O}-\text{H}$ bending vibration of water or the Amide I band rather than a $\text{C}-\text{H}$ bond, exhibited the highest intensity for the G3 filtrate (Fig. 6). This increase in the water-related signal relative to the non-polar grease components suggests effective hydrophobic pollutant removal, leading to a more hydrophilic effluent.

Absence of Aliphatic $\text{C}-\text{H}$ Stretches where Key aliphatic $\text{C}-\text{H}$ stretching peaks at 2922 cm^{-1} and 2853 cm^{-1} , and the CH_2 bend at 1460 cm^{-1} , prominent in the unfiltered grease sample (G0), were absent in all filtered samples (G1, G2, G3, G4), as indicated in Table 2 (see Table 1). The CH_3 bending peak at 1377 cm^{-1} was also absent post-filtration. The Amide I band (around 1634 cm^{-1}) and the $\text{O}-\text{H}$ stretch (around 3381 cm^{-1}) showed increased intensity after filtration across all filter types.

3.3. Efficacy of filter materials

Based on the ATR-FTIR spectral changes, particularly the disappearance of characteristic O&G functional groups and the relative intensities of remaining or newly formed bands, a qualitative assessment of filter efficacy was made. For oil removal from restaurant wastewater, the M3 filter combination (coconut coir + recycled beach sand + activated charcoal) was identified as the most effective. This conclusion was supported by the highest intensity of hydroxyl groups in the M3 filtrate. For grease removal from workshop wastewater, the G4 filter combination (coconut coir + recycled beach sand + inactivated charcoal) was determined to be the most effective, also correlated with the highest intensity of hydroxyl groups in its filtrate.

4. Discussion

The results obtained from the ATR-FTIR spectroscopy provide compelling evidence for the potential efficacy of the selected filter materials in removing oil and grease from restaurant and workshop wastewater, respectively. The discussion will focus on interpreting these

Table 1

Summary of ATR-FTIR functional group changes in restaurant wastewater (oil-laden) before and after filtration with different filter material combinations.

Types of functional group existed in samples before and after filtered	Condition of functional group before filtered	Condition of functional group after filtration
$=\text{C}-\text{H}$ (<i>cis</i> -) stretch (3006 cm^{-1})	Present (Sharp)	Presence (Broadened become $\text{O}-\text{H}$ group)
$\text{C}-\text{H}$ stretch (2872 cm^{-1})	Present (Strong)	Absence
$\text{C}=\text{O}$ stretch (1744 cm^{-1})	Present (Strong)	Absence
$=\text{C}-\text{H}$ (<i>cis</i> -) stretch (1418 cm^{-1})	Presence	Presence (Decreased)
CH_3 bending rocking (1377 cm^{-1})	Present (Moderate)	Absence
$\text{O}-\text{H}$ (3318 cm^{-1})	Absence	Emergence of Strong, Broad Peak
$\text{C}-\text{O}$ from environment ($2101, 2123, 2131, 2105\text{ cm}^{-1}$)	Absence	Presence
$-\text{CH}_2$ or CH_3 bond overlap with NH_2	Absence	Presence

Table 2

Summary of ATR-FTIR functional group changes in workshop wastewater (grease-laden) before and after filtration with different filter material combinations.

Types of functional group existed in samples before and after filtered	Condition of functional group before filtered	Condition of functional group after filtration
OH stretch (3381 cm^{-1})	Present (Broad)	Significant Intensity Increase
$\text{C}-\text{H}$ stretch (2853 cm^{-1})	Present (Strong)	Absence
$\text{C}-\text{H}$ stretch (2922 cm^{-1})	Present (Strong)	Absence
Amide I band due to $\text{C}=\text{O}$ stretching vibration of amide group (1634 cm^{-1})	Present (Moderate)	Presence (Slight Increase)
CH_2 bend (1460 cm^{-1})	Present (Moderate)	Absence
CH_3 bending rocking (1377 cm^{-1})	Present (Weak)	Absence

spectral changes, elucidating the role of different filter components, and contextualizing the findings within the broader goals of sustainable wastewater treatment.

4.1. Interpretation of functional group changes and removal mechanisms

Prior to analysing the adsorption mechanisms, it is essential to establish the distinct baseline chemical profiles of the two real-world effluents targeted in this study. Restaurant wastewater is predominantly laden with organic fats, oils, and grease (FOG), characterised chemically by triglycerides and free fatty acids. This theoretical profile perfectly aligns with our empirical observation of the strong ester carbonyl ($\text{C}=\text{O}$) peak at 1744 cm^{-1} in the unfiltered M0 sample. Conversely, automotive workshop effluent is dominated by synthetic hydrocarbons and metallic soap thickeners (such as lithium 12-hydroxystearate). This is precisely reflected in the G0 spectrum via the intense aliphatic $\text{C}-\text{H}$ stretches (2922 cm^{-1} and 2853 cm^{-1}) and the Amide I/ carboxylate interactions (1634 cm^{-1}). By establishing these distinct influent profiles, the subsequent disappearance of these specific peaks post-filtration directly validates the targeted efficacy of the composite materials.

In regards to restaurant wastewater, The complete absence of the ester carbonyl ($\text{C}=\text{O}$) peak at 1744 cm^{-1} in all filtered samples (M1-M4) is a crucial finding (Fig. 5). This peak is a hallmark of triglycerides, the primary constituents of vegetable and animal oils [16]. Its disappearance, along with the removal of aliphatic $\text{C}-\text{H}$ stretches, strongly indicates the effective adsorption of these oily compounds onto the filter media. Although ATR-FTIR spectroscopy effectively identifies functional group transformations, it remains a qualitative technique. Quantifying the absolute concentration of pollutants requires extensive multipoint calibration against standard gravimetric methods, which exceeds the qualitative scope of this initial pilot investigation.

The appearance of minor, sharp peaks around 2360 cm^{-1} in some filtered spectra is attributed to atmospheric CO_2 in the spectrometer's light path and is not a product of chemical reaction. Similarly, the absence of the $\text{C}-\text{H}$ stretch at 2872 cm^{-1} and CH_3 bending at 1377 cm^{-1} , both associated with aliphatic chains in lipids, further supports the removal of oil components. The emergence or increased intensity of broad $\text{O}-\text{H}$ bands ($3000\text{--}3700\text{ cm}^{-1}$) in filtered samples, particularly M3, suggests either the introduction of water molecules that were previously masked or the formation of hydroxyl-containing compounds as a result of interaction with the filter media, or simply a cleaner water matrix allowing for better detection of water's own $\text{O}-\text{H}$ signal (see Fig. 2).

To isolate the role of each filter component, a step-wise comparison of our stacked spectra (Fig. 3) is required. The M1 filter (coconut coir alone) achieved an initial reduction of the 1744 cm^{-1} ester peak, confirming its role in primary physical sorption and bulk oil retention. The addition of sand in the M2 composite provided further physical

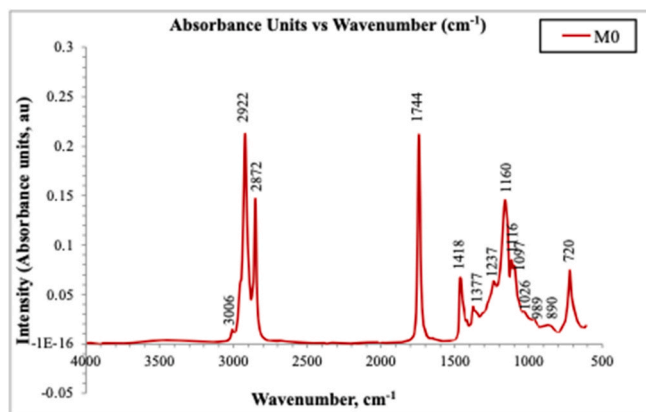


Fig. 2. Average ATR-FTIR spectrum of unfiltered restaurant wastewater (M0). The spectrum shows characteristic peaks of oil, including the C–H asymmetric and symmetric stretches of aliphatic chains at 2922 cm^{-1} and 2855 cm^{-1} , and a strong ester carbonyl (C=O) peak from triglycerides at 1744 cm^{-1} .

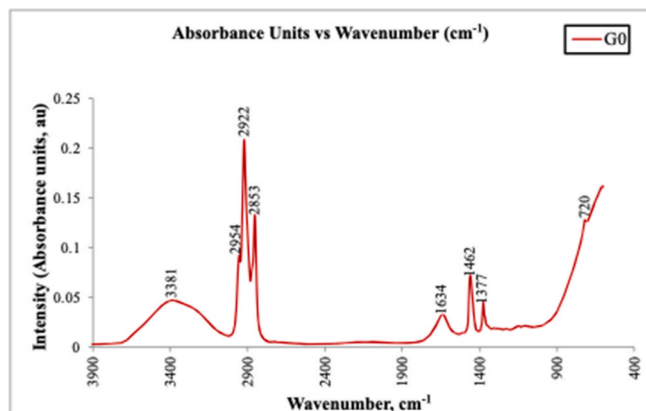


Fig. 3. Average ATR-FTIR spectrum of unfiltered workshop wastewater (G0). The spectrum displays characteristic peaks of grease, including a broad O–H stretch around 3381 cm^{-1} , strong aliphatic C–H stretches between 2954 and 2853 cm^{-1} , and an Amide I band at 1634 cm^{-1} .

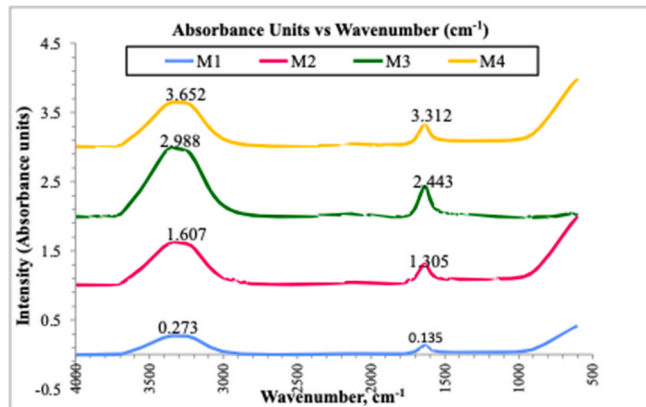


Fig. 4. Stacked ATR-FTIR spectra of restaurant wastewater after filtration through different material combinations: M1 (coconut coir only), M2 (coconut coir + recycled beach sand), M3 (coconut coir + recycled beach sand + activated charcoal), and M4 (coconut coir + recycled beach sand + inactivated charcoal).

straining, evidenced by the continued suppression of aliphatic chains. However, the complete eradication of residual hydrophobic traces and

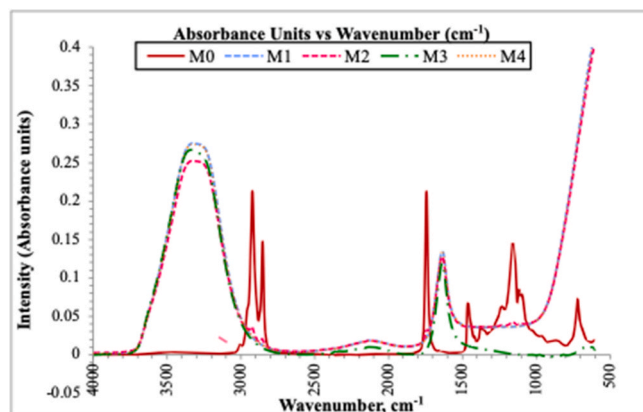


Fig. 5. Comparative ATR-FTIR spectra of restaurant wastewater before (M0) and after filtration with various filter material combinations (M1, M2, M3, M4), illustrating changes in oil composition.

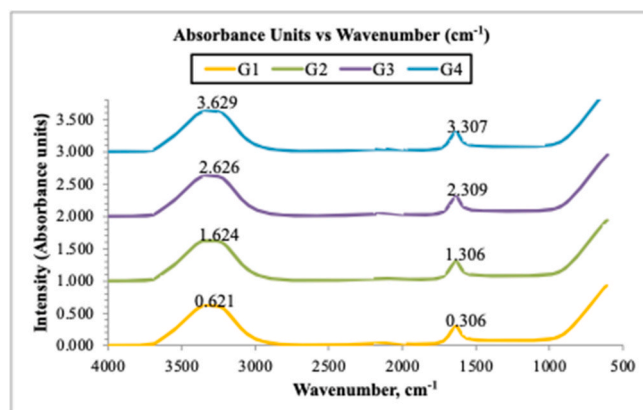


Fig. 6. Stacked ATR-FTIR spectra of workshop wastewater after filtration through different material combinations: G1 (coconut coir only), G2 (coconut coir + recycled beach sand), G3 (coconut coir + recycled beach sand + activated charcoal), and G4 (coconut coir + recycled beach sand + inactivated charcoal).

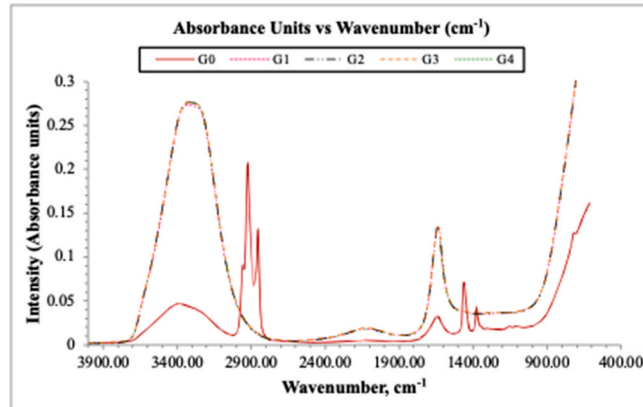


Fig. 7. Comparative ATR-FTIR spectra of workshop wastewater before (G0) and after filtration with various filter material combinations (G1, G2, G3, G4), illustrating changes in grease composition.

the optimal enhancement of the O–H stretch at 3318 cm^{-1} were only achieved with the inclusion of activated charcoal in the M3 composite. Coconut coir acts as an initial coarse filter and adsorbent due to its fibrous and porous nature [7,23,24]. This sequential spectral data

confirms that while coir and sand handle macro-filtration, the micro-porous structure of activated charcoal is strictly necessary for the final adsorption of smaller lipid molecules. The higher intensity of hydroxyl groups in M3 filtrate might reflect a more efficient removal of hydrophobic oil, leading to a relatively more hydrophilic (water-rich) effluent (see Fig. 4).

For workshop wastewater, the disappearance of the strong aliphatic C–H stretching vibrations at 2922 cm^{-1} and 2853 cm^{-1} (characteristic of methyl and methylene groups in grease components like 12-hydroxystearate [21] and the CH_2 bend at 1460 cm^{-1} across all filtered samples (G1–G4) signifies effective grease removal (Fig. 7). The increased intensity of the O–H stretch (around 3381 cm^{-1}) and the Amide I band (around 1634 cm^{-1}) post-filtration is noteworthy. The enhanced O–H signal in G4 (coconut coir + recycled sand + inactivated charcoal) again points to its superior efficacy. Inactivated charcoal, while potentially having a lower specific surface area than activated charcoal, can still possess significant adsorptive capabilities for larger organic molecules often found in grease (Fig. 6). The combination in G4 likely provides a multi-modal removal mechanism: physical straining by coir and sand, followed by adsorption onto the inactivated charcoal. The reason for inactivated charcoal outperforming activated charcoal (G3) for grease in this specific instance warrants further investigation but could be related to the specific molecular size distribution and polarity of the grease components, which might favor the pore structure or surface chemistry of the inactivated charcoal used.

4.2. Significance of hydroxyl group intensity and adsorption mechanisms

The consistent observation that the most effective filters (M3 for oil, G4 for grease) yielded filtrates with the highest hydroxyl (O–H) group intensity is a key finding. This is not interpreted as a quantitative measure of removal efficiency but rather as an indicator of a significant shift in the bulk sample's chemistry.

The removal of hydrophobic O&G components makes the remaining liquid matrix relatively richer in water, allowing the strong, broad O–H signal of water ($\sim 3400\text{ cm}^{-1}$) to dominate the spectrum. This effect, which can mask residual pollutant signals, is consistent with successful filtration.

The superior performance of G4 (with inactivated charcoal) for grease removal, compared to G3 (with activated charcoal), warrants explanation. While activated charcoal has a high surface area dominated by micropores ideal for small organic molecules in oil, grease consists of larger, more complex molecules (e.g., long-chain fatty acid salts).

It is plausible that the inactivated charcoal, having undergone less aggressive processing, retained a greater proportion of meso- and macropores. These larger pores may be more effective at physically trapping bulky grease molecules. Furthermore, its surface may be less polar, favoring adsorption of the long, non-polar hydrocarbon tails of grease components.

4.3. Contribution of individual filter materials

The study design allowed for an implicit understanding of the contribution of each material. Coconut coir, as the base material in all filters, likely provides initial sorption and physical filtration [11]. The addition of recycled sand enhances physical separation. The distinct roles of activated charcoal (in M3) for oil and inactivated charcoal (in G4) for grease highlight the importance of selecting appropriate adsorbent types based on the target pollutant's characteristics. Activated charcoal's efficacy for smaller organic molecules in oil aligns with its known properties, while inactivated charcoal's performance for potentially larger, more complex grease molecules suggests a different optimal adsorbent interaction.

4.4. Potential industrial applications and oil spill remediation

The findings of this study have significant implications for various industrial sectors that generate O&G-laden wastewater. The food processing industry, including slaughterhouses, dairy plants, and edible oil refineries, often discharges effluents with high O&G content. The low-cost and effective filter combinations developed here, particularly M3, could be adapted for pre-treatment or polishing stages in these facilities, reducing the load on downstream treatment processes and helping industries meet stringent discharge regulations. Similarly, automotive repair and maintenance workshops, metal fabrication units, and car washes, which contribute significantly to grease and hydrocarbon pollution, could implement systems based on the G4 filter combination for on-site treatment. The modular design of the filtration prototype also lends itself to scalability, allowing for adaptation to different volumes of wastewater.

Furthermore, the high adsorption capacity demonstrated by these natural and waste-derived materials, especially coconut coir and charcoal, suggests their potential utility in emergency response scenarios such as oil spills. Coconut coir is known for its oleophilic and hydrophobic properties, making it an effective sorbent for spilled oil on water surfaces or contaminated shorelines [24,25]. The materials investigated could be incorporated into booms, mats, or permeable reactive barriers for containment and cleanup operations. While this study focused on wastewater drains, the fundamental adsorption mechanisms are relevant. For oil spill applications, the materials might be deployed in a less structured manner than the column filters, but their inherent sorptive qualities for hydrocarbons remain advantageous. Further research would be needed to optimize their application for large-scale environmental remediation, focusing on deployment strategies, recovery of saturated sorbents, and their ultimate disposal or regeneration.

4.5. Limitations and future directions

This study successfully demonstrated the potential of these filter combinations using a qualitative, spectroscopic approach. However, several limitations must be acknowledged, which define a clear path for future research. While this study demonstrates the potential of these filter combinations, a quantitative analysis of O&G concentration (e.g., mg/L) before and after filtration would provide a more definitive measure of removal efficiency. The following limitations define a clear path for future research where, the primary limitation is the qualitative nature of the analysis. While ATR-FTIR demonstrates the disappearance of functional groups, it does not quantify removal efficiency. Future work must complement this spectroscopic data with standard quantitative methods, such as gravimetric analysis (mg/L), Chemical Oxygen Demand (COD), and Total Petroleum Hydrocarbons (TPH) analysis. Correlating spectral changes with these metrics will provide definitive performance data and validate the efficacy claims. Secondly, the study was conducted under gravity flow without controlling for flow rate or hydraulic retention time. These are critical parameters that heavily influence filter performance and longevity. Future experiments should systematically investigate the effect of varying flow rates and O&G loading concentrations to optimize the filter design for real-world applications. Third, for economic feasibility and sustainability, the regeneration and reusability of the filter media are paramount. This study did not explore regeneration. Future research should investigate methods such as solvent washing, thermal desorption, or backwashing to restore the adsorptive capacity of the filters. A full lifecycle assessment would be crucial to determine the overall environmental and economic viability of these systems. Fourth, the modular prototypes were tested at a laboratory scale. To assess true industrial applicability, pilot-scale studies under real-world operational conditions are necessary. This would involve constructing larger systems to handle higher volumes and continuous flows typical of restaurant or workshop effluents, allowing for an evaluation of performance, durability, and

maintenance requirements over extended periods. Finally, A deeper characterization of the adsorbents, particularly the activated and inactivated charcoal (e.g., using BET for surface area and porosity analysis), would provide a more robust explanation for their differential performance against oil and grease.

5. Conclusion

This study effectively demonstrated the potential of utilising locally sourced and low-cost materials—coconut coir, recycled beach sand, activated charcoal, and inactivated charcoal—in various combinations for the removal of oil and grease from restaurant and workshop wastewater, respectively. ATR-FTIR spectroscopy confirmed the significant reduction or absence of characteristic functional groups associated with O&G components in the filtered effluents, validating the qualitative efficacy of the developed filtration prototypes.

The filter combination M3 (coconut coir, recycled beach sand, and activated charcoal) proved to be the most effective for removing oil from restaurant wastewater. For grease removal from workshop wastewater, the G4 combination (coconut coir, recycled beach sand, and inactivated charcoal) exhibited superior performance. The enhanced intensity of hydroxyl (O–H) groups in the filtrates from these optimal combinations further corroborated their higher efficiency in producing cleaner water.

The findings highlight the synergistic effects of combining different filter materials, leveraging both physical filtration and adsorption mechanisms. This research contributes to the development of sustainable and economically viable wastewater treatment solutions, particularly relevant for decentralized applications in small to medium-scale enterprises and communities. The utilisation of waste-derived materials aligns with circular economy principles and contributes positively to achieving key Sustainable Development Goals related to clean water, responsible consumption, and environmental protection. While this qualitative study requires follow-up with quantitative analysis and pilot-scale testing, this study lays a strong foundation for the practical application of these natural filter systems in mitigating O&G pollution.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI in order to suggest formatting and structural changes to align the manuscript with specific journal guidelines. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRedit authorship contribution statement

Maliki Hapani: Conceptualization, Methodology, Supervision, Validation. **Nur Fatien Muhamad Salleh:** Data curation, Formal analysis, Investigation, Writing – original draft. **Khairilmizal Samsudin:** Data curation, Validation, Writing – review & editing. **Izzah Aqilah Puad:** Data curation, Investigation, Writing – original draft. **Baharuddin Zakaria:** Investigation, Methodology, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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